Fiscal Year:	FY 2015	**Task Last Updated:**	FY 03/23/2015
PI Name:	Fernandez-Pello, Carlos Ph.D.		
Project Title:	Material Ignition and Suppression Test (MIST) in Space Exploration Atmospheres		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Combustion science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	ferpello@me.berkeley.edu	**Fax:**	FY 510 642-6163
PI Organization Type:	UNIVERSITY	**Phone:**	510 642-6554
Organization Name:	University of California, Berkeley		
PI Address 1:	Department of Mechanical Engineering		
PI Address 2:	6105A Etcheverry Hall		
PI Web Page:			
City:	Berkeley	**State:**	CA
Zip Code:	94720-1740	**Congressional District:**	9
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2009 Combustion Science NNH09ZTT001N
Start Date:	05/21/2013	**End Date:**	10/31/2015
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA GRC
Contact Monitor:	Niehaus, Justin	**Contact Phone:**	(216) 433-3564
Contact Email:	justin.niehaus@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: Extended to 10/31/2015 per NSSC information (Ed., 3/24/15)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNX13AL10A		
Performance Goal No.:			
Performance Goal Text:			
	NOTE: Continuation of NNX10AE01G, Material Ignition and Suppression in Space Exploration Atmospheres. Central Objectives: The next generation of NASA spacecraft will use lower pressures and increased oxygen concentrations than current designs. Although there is insufficient knowledge about fire behavior of materials in these Space Exploration Atmospheres (SEA), recent research suggests that material ignitability is increased in SEA. These results imply that next generation spacecraft present a greater fire safety hazard than current designs. Although the most effective fire safety strategy is to prevent ignition, if ignition does occur, it is critical to rapidly extinguish the fire. Therefore, an experimental investigation into ease of ignition and fire suppression characteristics of materials under SEA is proposed.		

Task Description:	Methods/techniques: The proposed experimental apparatus is simple and easily incorporated in the Combustion Integrated Rack on the International Space Station (ISS). It consists of a small-scale cylindrical flow duct that has fuel samples attached to its inside wall. A cylindrical radiant heater is located at the center of the duct. An oxidizer gas flows at down the length of the tube. Electrical igniters are placed downstream of each fuel sample. Time to ignition and the time between introduction of suppression agent and extinction are measured as a function of environmental conditions. The normal gravity and microgravity experimental programs will be supported by a comprehensive model of piloted ignition and extinction of materials in SEA. Microgravity experiments will be used to validate the model, which in turn will be used to predict the fire behavior of different combustible materials in SEA. Perceived Significance: Currently, there exists no testing methodology specifically designed to determine the fire hazards of materials under those conditions. The proposed work will fill that void and provide additional information about the effect of SEA on the flammability and suppression of materials. It will also provide guidance to interpret and extend current NASA testing procedures to SEA.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The project is intended to enhance the ability to predict ignition, suppression, and spread characteristics of materials in their end use configuration. This approach is called performance based design in terrestrial fire protection engineering. Computational support for this approach requires detailed information about material thermal and combustion properties. Our work on MIST will use experimental data to validate computer models attempting to predict fire behavior using material property estimation techniques. These techniques are currently a focus of research in fire protection engineering.
Task Progress:	The Material Ignition and Suppression Test (MIST) project has had the objective of evaluating the ease of ignition and fire suppression of materials to be used in future manned spacecraft under the anticipated environmental condition. For this purpose, an experimental, numerical, and theoretical research program is being conducted on the effect of space exploration atmospheres (SEA) on the piloted ignition of representative combustible materials, their steady state flame spread characteristics, and on their fire suppression characteristics. The experimental apparatus and test methodology are derived from that of the Forced Ignition and Flame Spread Test (FIST), a well-developed bench scale test designed to extract material properties relevant to predictions of material flammability. In the FIST test, materials are exposed to an external radiant heat flux and the ignition delay and critical mass flux at ignition are determined as a function of the type of material and environmental conditions present. In the original MIST design, a small-scale cylindrical flow duct with fuel samples attached to its inside wall was heated by a cylindrical infra-red heater located along the central axis of the cylinder. However, as the project evolved it was decided by NASA that it would be better to produce an experimental design that could accommodate other experiments and experimenters with different concepts or sample geometries. Based on those instructions and input regarding the requirements of other researchers that may share the hardware in an ISS/CIR experiment, a cylindrical design placing the sample at the center of an optically transparent tube with heaters equally spaced along the exterior of the cylinder wall was developed. A hot wire igniter in the downstream boundary layer of the fuel sample initiates piloted ignition. Environment variables that can be studied via this experimental apparatus include: external radiant heat flux, flow oxygen concentration, flow velocity, ambient pressure, and gravity level (if flown in the ISS/CIR). This new geometry experimental design maintains good consistency with Dr. Tien's and Dr. Olson's project approaches and experimental objectives. A further goal of the project has been to develop a combined solid/gas phase numerical model based on the MIST test methodology to predict the flammability behavior of practical materials in spacecraft. Additionally, with the current flight of the Burning and Suppression of Solids – II (BASS – II) project aboard the ISS, work being done to accommodate the ground-based experiments associated with that project in the MIST experimental apparatus. MIST has developed an usable experimental apparatus for evaluation of an optimum configuration of sample, heater, and igniter placement to fall within the power supply and heat dissipation requirements of the NASA ISS/CIR. Using this experimental apparatus, time to ignition, and mass loss at ignition data have been collected as a means of establishing a baseline for eventual comparison with space flight data. In a parallel effort a 2-D axisymmetric numerical model of the piloted ignition of a sample has been developed and prediction of ignition delay is being obtained as a function of several ambient variables. In parallel, work is undergoing to employ this computational model to predict and extend the results of the BASS – II standard and microgravity experiments. For the ground-based (1g) MIST tests, fame spread rates were found for four diameters (0.3175, 0.635, 0.9525, 1.27 cm) of clear and black cast Poly(methyl-methacrylate) (PMMA) rods under four different external radiant heat fluxes (0, 2, 5, 10 kW/m2). It was found that there is a clear decreasing trend in flame-spread rate with increasing PMMA rod diameter. Similarly, for both clear and black cast PMMA rods and at both 19 and 21% oxygen by volume increasing the external radiant flux acts to increase the flame spread-rate. Comparing flame-spread rates from 19 and 21% oxygen concentration (1g) experiments for both clear and black cast PMMA rods, the flame-spread rates are almost a magnitude higher for the 21% oxygen environment than for the 19% oxygen environment. It seems though, that the effect of oxygen concentration between the 19% and 21% environments is magnified by the clear cast PMMA rods compared to the results in the black cast PMMA rods. As a part of the BASS-II campaign, tests were recently carried out in the Microgravity Science Glovebox (MSG) aboard the International Space Station (ISS). Three different diameters (0.635, 0.9525, 1.27 cm) of clear and black cast PMMA rods were tested. Oxygen concentration was varied between 16% and 21% by volume, and flow velocities were varied between 0.4 and 8 cm/sec. Tests were conducted at fixed oxygen concentration, and flow velocity was varied downward towards extinction. Across all three diameters, there seems to be a correlation between decreasing oxygen concentration and decreasing flame spread rate, as well as a trend in decreasing flame-spread rate with increasing diameter as was seen in the 1g experimental data. Additionally, there seems to be a mild trend of increasing flame-spread rate with increasing opposed flow velocity. For all three diameters for which tests were conducted in both μg and 1g the flame spread rates are higher in the case of the microgravity tests. Additionally, as previously discussed, we see decreased flame-spread rates as the diameter of the rods increases.

Bibliography Type:	Description: (Last Updated: 02/15/2026)
Abstracts for Journals and Proceedings	Link S, Fernandez-Pello C, Olson S, Ferkul P. "Piloted Ignition and Flame Spread Over Clear and Black PMMA Cylinders in Opposed Flow." 30th Annual Meeting of the American Society for Gravitational and Space Research, Pasadena, CA, October 22-26, 2014. 30th Annual Meeting of the American Society for Gravitational and Space Research, Pasadena, CA, October 22-26, 2014. , Oct-2014
Abstracts for Journals and Proceedings	Link S, Fernandez-Pello C. "Piloted Ignition and Flame Spread of Clear and Black PMMA Cylinders with an External Radiant Heat Flux." Poster presented at the 35th International Symposium on Combustion, San Francisco, CA, August 3-8, 2014. Program and abstracts. 35th International Symposium on Combustion, San Francisco, CA, August 3-8, 2014. Abstract W2P061. , Aug-2014
Articles in Peer-reviewed Journals	Fereres S, Fernandez-Pello C, Urban DL, Ruff GA. "Identifying the roles of reduced gravity and pressure on the piloted ignition of solid combustibles." Combustion & Flame. 2015 Apr;162(4):1136-43. http://dx.doi.org/10.1016/j.combustflame.2014.10.004 , Apr-2015